Yungyeom An

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RESEARCH INTERESTS

Design of functional soft materials; Molecular architecture and topology; Structure-property relationships in soft matter; Supramolecular chemistry; Molecular interactions and self-assembly; Soft matter physics; Experiment-simulation integration

EDUCATION

Daegu Il Science High School, Korea | High School Diploma — 2018.02 – 2020.02

Pohang University of Science and Technology (POSTECH), Korea | B.Sc., Chemistry — 2020.02 – 2025.02
GPA: 4.08/4.3 (U.S. Equivalent: 3.92/4.0) – ranked 1st in class

TU Wien, Austria | Exchange student — 2023.10 – 2024.02

Nanyang Technological University, Singapore | Summer exchange student — 2024.06 – 2024.07

RESEARCH EXPERIENCE

Center for Self-assembly and Complexity (Prof. Kimoon Kim) — POSTECH, Korea
Undergraduate Researcher — 2022.06 – 2023.02; Resumed 2025.10 – present

- Presented a poster at the 130th General Meeting of the Korean Chemical Society (KCS).
- Led a project on programmable spatiotemporal control of chemical processes via multi-frequency acoustic stimulation for out-of-equilibrium polymer materials (manuscript in preparation).
- Investigated microgel-based polymer systems to develop pH-responsive interfacial patterning.

Smart Soft Materials Lab (Prof. Youn Soo Kim) — POSTECH, Korea
Undergraduate Researcher — 2024.09 – 2024.12

- Synthesized lignin-based hydrogels by engineering cation–π interactions and conducted mechanical (rheology, tensile, lap shear) and chemical characterization to optimize for bio-adhesive applications.
- Developed skills in polymer formulation, crosslinking control, and assessment of structure-property relationships in soft-matter materials.

Max Planck Institute for Polymer Research (Prof. Mischa Bonn) — Mainz, Germany
Intern (Korea-Germany Junior Research Fellowship Program) — 2023.08 – 2024.05

- Collaborated with international researchers and participated in scientific discussions at VeKNI, strengthening cross-disciplinary scientific communication skills.
- Elucidated the hydrogen bond asymmetry and anticorrelation in liquid water by integrating MD simulations (GROMACS) with 2D-IR spectroscopic data, successfully reproducing water's structural anomalies.
- Analyzed temperature-dependent IR spectroscopy and improved data processing workflows (MATLAB/Origin) for reproducible analysis.

Computational Energy/Bio Soft Material Lab (Prof. Chang Yun Son) — POSTECH, Korea
Undergraduate Researcher — *2023.02 – 2023.06*

- Performed MD simulations (OpenMM) to investigate image-charge effects on ionic liquid systems (EMIM/BF4 and BMIM/BF4) confined in graphene electrodes, revealing enhanced charge separation at varying potentials.
- Gained proficiency in atomistic simulation techniques (WHAM/Umbrella Sampling for PMF calculations, protein folding simulations, and water-model validation).

ADDITIONAL RESEARCH TRAININGS

Quantum Dot Bioconjugation for Targeted Imaging — POSTECH, Korea
Winter Research Program — 2022.01 – 2022.02

- Synthesized antibody-conjugated core-shell quantum dots (CdSe/ZnS, AgInS/ZnS) and developed surface functionalization strategies for targeted bio-imaging.

Functional Polymer Thin Films & Bio-interfaces KAIST, Korea

Technical Training 2025.08 – 2025.10

- Trained in iCVD fabrication of polymer nanocoatings, utilizing FTIR and water contact angle for characterization, and evaluating bio-interfacial properties via *in vitro* assays (cell culture, PCR, and immunofluorescence staining).

TEACHING

Student Mentoring Program | POSTECH 2021.03 – 2023.06

- Taught General Chemistry and Organic Chemistry to freshmen and sophomores, and mentored academic planning.

LEADERSHIP

The POSTECH Times | POSTECH 2020.04 – 2023.03

Editor-in-chief (2022.08 – 2023.03); Head of the Reporting Office (2022.02 – 2022.07); Reporter

- Oversaw monthly publication as Editor-in-Chief; managed editorial planning and supervised reporting teams.
- Led an exclusive interview with a Nobel laureate and edited scientific feature articles for the university community.
- Strengthened organizational leadership through team management and cross-department coordination.

Management Strategy Student Association (MSSA) | POSTECH 2021.03 – 2021.12

- Collaborated on case analyses of technology/entertainment companies, enhancing analytical and strategic thinking.
- Mentored new members to improve presentation and communication skills.

Chemistry Student Council | POSTECH 2022.01 – 2022.12

Treasurer

- Managed a $6,000 departmental budget and organized welfare and academic events for 90 students.
- Facilitated a career exploration event by coordinating faculty–student discussions.

Student Council (Entire Student Body) | POSTECH 2020.03 – 2021.07

Financial department

- Managed budgeting and accounting for the university-wide Student Council through cross-departmental collaboration.
- Organized community-building events and supported student engagement during the pandemic.

HONORS & AWARDS

Jigok Scholarship (Full Tuition Scholarship) | POSTECH 2020.02 – 2024.02
Chemistry Scholarship of Eminent Members | Chemistry department, POSTECH 2021.09
Lee In-sook Scholarship (Top Academic Excellence Scholarship) | Chemistry department, POSTECH 2022.07
Research Support Fund | Max Planck POSTECH/Korea Research Initiative 2023.08 – 2024.01
MIRAE ASSET Global Exchange Scholarship | Mirae Asset Park Hyeon Joo Foundation 2023.07
Jung Sung-gi Award (Graduation with Highest Honor) | Chemistry department, POSTECH 2025.02
President's Award | Korean Society for Biochemistry and Molecular Biology 2025.02

SKILLS

Languages	English (proficient); Korean (native); German (elementary)
Programming	Python; C; MATLAB; Linux (basic); VMD
Techniques	MD simulation (GROMACS, OpenMM; WHAM, Umbrella Sampling)
	Spectroscopy (IR, UV-Vis, PL, NMR)
	Materials (rheometry, mechanical testing, hydrogel synthesis, iCVD; bioconjugation, staining)

REFERENCES

Mischa Bonn | Director, Max Planck Institute for Polymer Research, Mainz, Germany
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Kimoon Kim | Department of Chemistry, POSTECH, Pohang, Republic of Korea
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Chang Yun Son | Assistant Professor, Department of Chemistry, Seoul National University, Seoul, Republic of Korea
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